R Basics

Lecture 1

Dr. Eric Friedlander

Before we start...

- Make sure you have done the following:
  - Downloaded and installed R
  - Downloaded and installed Positron (or R Studio)
  - Created a Gemini Pro AI account (free for students)

Introduction to R

What is R?

- R is a statistical programming language
  - Inspired by the S programming language developed at Bell Labs in the 1970s
  - Created by Ross Ihaka and Robert Gentleman at the University of Auckland, New Zealand in the early 1990s for teaching statistics
  - Now one of the most popular programming languages for data science and statistics
- R is free and open source
  - Anyone can download and use R for free
  - Anyone can contribute to the development of R and its packages

R is not the same as R Studio, R Markdown, or Positron

- R is the programming language
- R Studio (now Positron) is an integrated development environment (IDE) for R
  - Provides a user-friendly interface for writing and running R code

R: Engine	RStudio: Dashboard
	

Figure 1: Source: Statistical Inference via Data Science

Working with R

Writing Scripts in R

- Let's open up a script and run some R code!
  - Running R from the console
  - Running R from a script file

R Packages

- R has a rich ecosystem of packages that extend its functionality
  - You can install packages using `install.packages("package_name")` (only do this once)
  - Load packages into your R session using `library(package_name)` (do this every time you start a new R session)
- An important package for this class:
  - `tidyverse`: A collection of packages for data manipulation, visualization, and modeling
    * `ggplot2`: A package for creating graphics using the Grammar of Graphics
    * `dplyr`: A package for data manipulation
- Let's install and load `tidyverse`

Palmer Penguins Dataset

- Install the `palmerpenguins` package
- Contains dataset on the famous Palmer Penguins: penguin species, island in Palmer Archipelago, size (flipper length, body mass, bill dimensions), and sex.
- Explore more with `?penguins`

Important Vocabulary

- **Data frame**: A table-like structure in R that holds data (`penguins`)
  - In the `tidyverse` data frames are referred to as `tibbles`
- **Variable**: is a quantity, quality, or property that you can measure (e.g. `species`, `bill_length_mm`, `island`)
- **Observation** or **Data point**: a single object that has been measured (e.g. one of the penguins in the data set)
- **Value**: the entry for a particular variable and observation (e.g. for the first penguin, the value of `species` is "Adelie", the value of `bill_length_mm` is 39.1)
- **Tabular Data**: data organized in rows and columns (like a spreadsheet or data frame)
- **Tidy Data**: tabular data where each observation is a row, each variable is a column, and each type of observational unit forms a table

Taking a quick look at `penguins`

- `glimpse` gives a quick overview of the data frame
- `head` shows the first few rows of the data frame
- `tail` shows the last few rows of the data frame

Wrap up

Do Next

1. Read the Introduction of R for Data Science
2. Post any question you have to the HW channel in our Team.
3. Move on to Lecture 02.